

Finite Spectral Shadows for the Collatz Valuation Cocycle

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Abstract

This document merges the theoretical note proposing the deterministic defect cocycle $\mathfrak{D}_{\text{Coll}}$ for the accelerated Collatz/Syracuse map with the results of a dedicated C numerical test suite built to probe it. Part I states the object and the conjectural *Quenched Inverse Principle* unchanged in substance from the original note, with the corrections suggested in review folded in as remarks. Parts II–IV report the suite: its construction, the per-experiment findings, and a dedicated analysis relating the empirically detected “shadow” windows to the finite transfer-operator gaps. The headline outcomes are (i) the load-bearing mod 3^r congruence identity is verified to $\sim 2.4 \times 10^7$ checks; (ii) the finite character-sector gap $|\hat{p}(\xi)| < 1$ is realized, but the *exact* coupled operator departs materially from the i.i.d. Fourier shadow; and (iii) single-orbit shadows correlate with the operator gap only at the level of the tilt (they concentrate at the critical tilt $s = 1$, where the gap is weakest), while the per-sector signal is confounded by finite residue-class count—a concrete instance of the “circularity leak.” Nothing here proves anything about Collatz; the purpose is to decide whether $\mathfrak{D}_{\text{Coll}}$ is the right finite detector and where it sees the obstruction most clearly.

Remark (Status). Part I builds on the pro-object $\mathfrak{Q}_{\text{Coll}}$ of the previous note (the quenched residue-height spectral shadow); project-internal terms (H23a, the QSD machinery, the residue spaces Ω_Q , the tilted operators $\mathcal{L}_{s,Q,L,\chi}$) are carried over without restatement. The *Quenched Inverse Principle* is conjectural. Throughout, $e(t) = e^{2\pi it}$. Empirical numbers in Parts III–IV come from the suite `collatz_tests` (builds FAST128/GMP); every figure is reproducible from the recorded run.

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Part I

The object and the principle

1 The obstacle that fixes the shape of the object

A single integer x_0 gives one deterministic valuation word $w(x_0) = (b_0, b_1, b_2, \dots)$, where $b_i = v_2(3x_i + 1)$. The annealed law is the pushforward of Haar measure on $\mathbb{Z}_2$ under $x \mapsto v_2(3x + 1)$. So “the orbit obeys the annealed law” means: the one prescribed word $w(x_0)$ equidistributes for the residue–height skew product. There is no ergodic theorem for a single prescribed non-generic point, and there cannot be: ergodicity yields almost-everywhere statements, which are silent about any individual orbit.

This dictates the shape of the missing object. It must be a *dichotomy with a closure*, not a measure-theoretic average: a pathwise quantity forced either to decay (the orbit equidistributes) or to localize as a finite, excludable obstruction. The quantity must reduce, at the trivial character and after ensemble averaging, to the annealed pressure $M(s)$ and the Cramér tilt; and its localization must land inside a fixed finite character sector, where H23a can act.

2 The object: a deterministic defect cocycle

Over each finite level (Q, L) , each tilt s , and each character $\chi \neq 1$ of Ω_Q , define the pathwise twisted Birkhoff functional

$$W_n(x; \chi, s) = \frac{1}{n} \sum_{i=0}^{n-1} \chi(x_i \bmod Q) 2^{s\Delta_{b_i}}, \quad \Delta_b = \log_2 3 - b.$$

The family of these functionals, fibered over Ω_{Coll} , is the object; call it the *deterministic defect cocycle* $\mathfrak{D}_{\text{Coll}}$. At $s = 0$ this is the residue Weyl sum: the orbit equidistributes modulo Q iff $W_n(x; \chi, 0) \rightarrow 0$ for every $\chi \neq 1$. The annealed pressure $M(s)$ and the critical tilt $p_b^{(1)} = 3 \cdot 4^{-b}$ are the $\chi = \mathbf{1}$, ensemble-averaged shadow of the same functional, so $\mathfrak{D}_{\text{Coll}}$ is the quenched lift of the thermodynamic formalism: $M(s)$ is what W_n self-averages to when nothing obstructs.

The content of $\mathfrak{D}_{\text{Coll}}$ at finite level is one statement in three languages—*equidistribution failure*, *affine/cohomological*, and *spectral*:

$$\begin{aligned} \limsup_n |W_n(x; \chi, s)| > 0 &\iff \chi \text{ is an approximate coboundary along the orbit} \\ &\iff \mathcal{L}_{s,Q,L,\chi} \text{ has a near-peripheral eigenstate read off from } \mu_n^x. \end{aligned}$$

Remark (Correction adopted from review). The three-way *equivalence* is too strong and is used here only in the safe direction. A non-decaying twisted sum signals correlation with χ , but need not be an exact coboundary (it could be concentration, subsequential trapping, nonstationarity); and producing a near-*top* eigenstate of $\mathcal{L}_{s,Q,L,\chi}$ is precisely the open keystone, not an automatic consequence. The load-bearing statement is the single implication: a persistent W_n bias $\implies$ a nontrivial finite character shadow of the orbit, and (conjecturally) a near-peripheral state of $\mathcal{L}_{s,Q,L,\chi}$.

3 Why the annealed side has a gap, explicitly

Use the exact affine cocycle

$$x_n = \frac{3^n x_0 + A_n}{2^{B_n}}, \quad A_n = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{B_j}, \quad B_n = \sum_{i < n} b_i.$$

Reducing modulo 3^r , for $n \geq r$ the $3^n x_0$ term vanishes and only the last r valuations survive.

Lemma 1 (Sliding-window congruence). *For $n \geq r$,*

$$x_n \equiv \sum_{k=1}^r 3^{k-1} 2^{-(b_{n-1} + \dots + b_{n-k})} \pmod{3^r}.$$

Since 2 is a primitive root modulo 3^r (order $2 \cdot 3^{r-1}$), the character value $\chi_3(x_n)$ is a sliding-window function of the partial sums $B_i \bmod 2 \cdot 3^{r-1}$. The single-symbol Fourier transform of the annealed law is

$$\hat{p}(\xi) = \sum_{b \geq 1} 2^{-b} e(b\xi) = \frac{e(\xi)}{2 - e(\xi)}, \quad |\hat{p}(\xi)| = \frac{1}{\sqrt{5 - 4 \cos 2\pi\xi}} < 1 \quad (\xi \notin \mathbb{Z}).$$

The frequencies attached to a nontrivial χ are $\xi = k/(2 \cdot 3^{r-1})$, all nonintegral, so to leading order in the i.i.d. model $\rho(\mathcal{L}_{0,Q,L,\chi}) = \max_{\xi} |\hat{p}(\xi_{\chi})| < 1$. **[Inference]**; this is the i.i.d. value, *not* the exact spectral radius of the coupled finite operator. The distinction is enforced throughout Parts III–IV: `rho_annealed` and `rho_exact` are reported in separate columns and never substituted.

4 The Quenched Inverse Principle

Quenched Inverse Principle (conjectural)

Let x_0 have an infinite forward orbit, and suppose it violates the annealed law persistently: for some **non-coboundary** χ on some Ω_Q , some tilt s , and some $\delta > 0$, $|W_n(x_0; \chi, s)| \geq \delta$ along a positive-density set of n (or, more weakly, along blocks of length $\ell_j \rightarrow \infty$). Then there exist a finite pair (Q, L) and a χ -sector eigenstate of $\mathcal{L}_{s,Q,L,\chi}$ with $|\lambda| \geq \rho(\mathcal{L}_{s,Q,L,0}) - o(1)$, exhibited by the orbit’s own empirical measure. Conjoined with the gap $\rho(\mathcal{L}_{s,Q,L,\chi}) \leq (1 - \kappa_Q) \rho(\mathcal{L}_{s,Q,L,0})$ for non-coboundary χ , this is a contradiction. Hence every integer orbit equidistributes relative to the annealed (valuation-pushforward) residue law.

Remark (Coboundary refinement, forced by the computations). The qualifier “**non-coboundary**” is essential and is the one substantive correction the numerics forced. The faithful operator has exactly one structural zero-gap sector—the quadratic character χ_2 —and it is *not* a defect but an exact coboundary: $\chi_2(x_i) = (-1)^{b_i-1}$ is valuation parity in disguise (appendix A), so the residues are not uniform mod 3 (they sit at $\frac{1}{3} : \frac{2}{3}$) and χ_2 belongs to the height/valuation *reference*. H23a is therefore not “all nontrivial characters are gapped” but

all *non-coboundary* characters are gapped.

After quotienting the (unique) coboundary, the gap is present, Q -uniform, and in fact constant (appendix A, appendix B). The positive-density hypothesis is also likely too strong—obstructions may be sparse—so the principle is stated to admit long blocks $\ell_j \rightarrow \infty$ (the form used in Exp. I).

5 The keystone and the two leaks

The single arrow that is not a transcription of known structure is

persistent quenched bias along $w(x_0) \implies$ a **finite-level** approximate eigenfunction at the **top** eigenvalue.

Finiteness leak. The limiting obstruction could live at $Q \rightarrow \infty$. Closing this needs an effective, uniform gap $\kappa_Q \gtrsim 1/\text{polylog}(Q)$, decaying more slowly than the entropy production of the bias. That quantitative H23a is a hypothesis.

Circularity leak. Reading a top eigenfunction off μ_n^x presumes the orbit’s residue frequencies are already near-stationary—what one is trying to prove. Breaking this needs an input not from the orbit’s own statistics; the best candidate (a regeneration-plus-bilinear engine resting on the multiplicative independence of 2 and 3, with a Bourgain–Sarnak–Ziegler step over the 3-adic frequencies) is **[Speculation]**, and is morally Sarnak-hard.

6 Three falsifiable consequences (the test targets)

1. For each r , compute the exact $\rho(\mathcal{L}_{0,3^r,L,\chi})$ and check it tracks $\max_k(5 - 4\cos(2\pi k/(2 \cdot 3^{r-1})))^{-1/2}$. A parametrically smaller coupled gap means the finiteness leak is real and returns the true κ_Q .
2. Along long orbits, compute the windowed $W_n(x; \chi, 0)$ at resonant frequencies; the prediction is decay with no persistent plateau. A plateau is the spectral shadow made visible and a direct test of the circularity leak.
3. At $s = 1$, compare $\rho(\mathcal{L}_{1,3^r,L,\chi})$ with $\rho(\mathcal{L}_{1,3^r,L,0})$. A nontrivial gap there is the statement that a high ascent cannot also hold a residue resonance.

Part II

The numerical test program

7 What was built

The suite `collatz_tests` (ISO C11; profiles FAST128 with `_uint128_t`, GMP with `mpz_t`, and a DEBUG profile of invariant assertions) implements nine experiments, A–I, over the accelerated step $S(x) = (3x + 1)/2^b$, $b = v_2(3x + 1)$. Modules cover the core arithmetic, a streaming orbit scanner, finite characters (additive $\chi_k(a) = e(ka/Q)$ and multiplicative on $(\mathbb{Z}/3^r)^\times$ with 2 a primitive root), the defect sums $W_{n,\ell}$ via complex prefix sums, empirical valuation/entropy statistics, an excursion/critical-event recorder, the finite residue–height operator builder, and a CSR sparse-matrix spectral toolkit.

Four design requirements, adopted from review, shaped the build:

1. the mod 3^r congruence (lemma 1) is treated as a *correctness gate* (Exp. H), verified before any spectral code relies on it;

2. ρ_{annealed} and ρ_{exact} are kept in separate columns and never interchanged (Exp. F);
3. the shadow search reports per-window magnitude *and* length, so both lim sup plateaus and long sparse blocks are visible—positive density is not required (Exp. I);
4. the quenched pressure is computed as $\limsup_n \frac{1}{n} \sum_{i < n} s \Delta_{b_i}$, not an ill-defined sum over words.

A definitional caveat carried throughout. The project-internal “ C_A -critical” certificate is not restated in the source notes. The spec’s literal inequality $V^2 \leq 2^A P$ (launch V , peak P) is *degenerate* in practice: once the orbit descends to small record-low launches, V^2 is tiny and essentially every excursion qualifies. The suite therefore offers two readings via `--crit-mode: vsq` (literal) and `height` (peak $\geq \text{launch} \cdot 2^A$, i.e. “rises A octaves”), with `height` the default. Both are reported; the formal certificate should be fixed before the $B/D/I$ “critical” labels are treated as canonical.

Part III

Findings

All runs below are from a single large sweep (FAST128 release).

8 Experiment H — the congruence gate

Verified

lemma 1 holds in every test: 23,905,968/23,905,968 checks match (FAST128, seeds 3–999,999, $r \leq 8$, 0 overflow), plus 144/144 under GMP for a 16-digit seed at $r = 12$ with n up to 155 (values $\gg 2^{128}$).

The load-bearing identity is sound; downstream residue/spectral code may rely on it. Each check compares the directly reduced orbit value $x_n \bmod 3^r$ against the sliding-window formula using $2^{-1} \bmod 3^r$.

9 Experiment A — height-tail census

Pre-registration: $\log_2 \Pr(h \geq H) \approx -H + C$. Over octaves 10–28 at 10^5 seeds/octave, the per-step slope of $\log_2(\text{tail}/\text{seeds})$ at octave 24 averages ≈ -0.97 for $H = 3, \dots, 11$ (steeper in the low-count tail). **Slope -1 confirmed.**

10 Experiment B — the Cramér-tilt law

Pre-registration: the bulk valuation law of high-ascent excursions tends to $p_b^{(1)} = 3 \cdot 4^{-b}$ with entropy-per-digit $\rightarrow H_2(3/4) = 0.8113$.

Cramér tilt confirmed (in direction)

Octave 26, 2×10^5 seeds, boundary-excluded *middle* segments of length ≥ 16 (103,976 segments): $\hat{p} = (0.657, 0.229, 0.089, 0.021)$ —between annealed (.5, .25, .125, .0625) and critical (.75, .1875, .0469, .0117)—with $D(\hat{p} \parallel p^{(1)}) = \mathbf{0.093} \ll D(\hat{p} \parallel p) = 0.191$ and entropy-per-digit 0.870.

Full segments dilute toward annealed because launch/exit boundary layers contaminate them—precisely what the boundary-layer test isolates. Larger scale roughly doubles the margin in favor of $p^{(1)}$ over the small run.

11 Experiment C — defect windows

At $s = 0$ (the residue Weyl sum) $|W|$ decreases with window length (0.31 at $L=16 \rightarrow 0.24 \rightarrow 0.20 \rightarrow 0.11$ at $L \approx 116$ for $Q = 27$), the equidistribution direction. Single-orbit windows stay short (orbits terminate at 1), so the asymptotic decorrelation lives in the operator experiments E/F rather than in single-orbit W .

12 Experiment D — post-exit taboo

Entropy-space repulsion present

Octave 24, 6,496 post-exit windows: $D(\hat{p}_{\text{post}} \parallel p) = 0.177 \ll D(\hat{p}_{\text{post}} \parallel p^{(1)}) = 0.576$, mean $M_{\text{post}}(1) = 0.989 < 1$ (53% subcritical). Post-exit relaxes toward annealed, away from the critical tilt.

This is the proposed reason two critical excursions fail to concatenate.

13 Experiments E and F — the finite operator gap

The killed residue–height operator $\mathcal{L}_{s,Q,L,\chi}$ is built with the forward residue map $a' = (3a + 1)2^{-b}$ for odd Q (so 2 is invertible, covering $Q = 3^r$), the height update $z' = z + \lfloor R(\log_2 3 - b) \rfloor$ with killing outside $[0, L]$, and the character phase on the source residue.

Target 3 (the $s = 1$ gap) and Target 1 (exact vs. annealed)

On the reliable *residue-only* operator (Exp. F), every nontrivial 3^r -sector is gapped, $\rho(\mathcal{L}_\chi) < \rho(\mathcal{L}_0)$, and the gap is *weakest at the critical tilt*: $s=1$ ratios run 0.54–0.93 versus 0.22–0.76 at $s=0$. Against the i.i.d. Fourier prediction $|\hat{p}(\xi)|$, the exact ratios deviate by ± 0.1 –0.36 with $\text{corr} = \mathbf{0.38}$ and mean $|\text{dev}| = 0.254$ across 48 sectors.

Remark (A solver defect surfaced by Part IV, since fixed). The *killed-height* operator (Exp. E) at first told a different story: its power iteration did not converge (relative residuals 3–9%; all sectors hit the 3000-iteration cap), because the height-transport mode produces a near-degenerate band of top eigenvalues, and several reported gap ratios *exceeded* 1—impossible, since with $|\mathcal{L}_\chi|$ equal to the principal operator on the same support Wielandt’s theorem forces $\rho(\mathcal{L}_\chi) \leq \rho(\mathcal{L}_0)$. The solver was

replaced by a **restarted complex Arnoldi** (Krylov subspace, Hessenberg shifted-QR Ritz values, validated against known spectra in the unit tests). The killed-height operator now converges to residual $< 8 \times 10^{-11}$ with no Wielandt violations.

Remark (The uniform gap is governed by the *worst* character). A uniform gap means $\rho(\mathcal{L}_\chi) \leq (1 - \kappa_Q)\rho(\mathcal{L}_0)$ for *all* $\chi \neq 1$, so $\kappa_Q = 1 - \max_\chi[\rho(\mathcal{L}_\chi)/\rho(\mathcal{L}_{\chi_0})]$ —the *worst*-gapped character. (Computed over the full group via the conjugate symmetry $k = 1..[\phi/2]$.)

Target 1 (the finiteness leak), full character group

Residue-only (the object that governs W): a uniform gap survives. $\kappa_Q = 1 - \max_\chi$ ratio *stabilizes* as Q grows—0.236 at $s=0$, 0.074 at $s=1$ for all $Q \in \{27, \dots, 2187\}$ —bounded well away from 0, with the worst character at a *fixed* frequency, not $k = 1$. The lowest-frequency $k=1$ char, where the annealed $1 - |\hat{p}(1/\phi)|$ collapses to 0 ($0.42 \rightarrow 2 \times 10^{-5}$ as $Q \rightarrow 2187$), is gapped by the *exact* operator by 0.24–0.78: the exact operator gaps low frequencies *far more* than the i.i.d. shadow. The annealed finiteness leak is **refuted** for the residue operator over $3^2 \rightarrow 3^7$.

Coupling the height to the character

The faithful *skew-product* killed-height operator (one b drives residue and height) has *no* uniform gap: its maximum gap ratio is *exactly* 1, at the **quadratic character** $k = \phi/2$. This is structural, not a discretization artifact—it stays exactly 1 across $R \in \{2, \dots, 16\}$ and $L \in \{24, \dots, 120\}$. The mechanism: the height is $\approx n \log_2 3 - B_n$, so it encodes B_n , and the quadratic character’s 2-adic factor $\chi(2)^{-b} = (-1)^b$ accumulates to $(-1)^{B_n}$ —a *coboundary* of the skew product, removable by a diagonal similarity for any height geometry.

The rebuilt coupling restores the gap

Decoupling the synchronization— $\mathcal{L} = \mathcal{L}_{\text{res}}(\chi) \otimes T_{\text{height}}$, the height a residue-independent survival factor, the character twisting only the residue factor (**--height-mode factored**)—gaps the quadratic character (ratio 0.333 at $s=0$, 0.600 at $s=1$, the residue value) and gives the killed operator the residue uniform gap exactly: $\kappa_Q = 0.345$ ($Q=9$) $\rightarrow$ 0.236 ($Q \geq 27$) at $s=0$, matching F. The Kronecker height factor cancels in every gap ratio, so this holds for all Q . The **killed operator now inherits the stabilizing residue gap and gaps every character**; the skew-product $\kappa_Q = 0$ is isolated to the genuine quadratic-character height-coboundary.

14 Experiment G — trace and determinant shadows

For small (Q, L) , $\text{tr}(\mathcal{L}^1) = 0$, $\text{tr}(\mathcal{L}^2) = 1.25$, $\text{tr}(\mathcal{L}^3) = 0$, $\text{tr}(\mathcal{L}^4) = 0.594$, and $\det(1 - z\mathcal{L})$ is smooth and positive (0.95 at $z=0.25 \rightarrow 0.82$ at $z=0.5$): no spurious peripheral structure.

15 Experiment I — the quenched spectral-shadow search

A window is flagged shadow-bearing when $\max_{Q,\chi,s}|W| \geq \delta$ and exceeds $3\times$ a length-scaled null $\text{rms}(w)/\sqrt{\ell}$ (the raw δ -only criterion saturates at 100% from short-window and max-over-grid inflation). Over 1,254,232 windows (octaves 18–24, moduli $\{9, 27, 81, 243\}$), 4.2% flag.

by label	windows	frac. shadow	by length	windows	frac. shadow
critical_C_A	61,372	0.084	8–15	331,996	0.003
high_ascent	445,193	0.054	16–31	330,930	0.029
post_exit	747,621	0.032	32–63	591,088	0.072
			64–127	218	1.000

Critical excursions carry shadows most often and post-exit least (taboo-consistent), and the shadow rate grows monotonically with window length, reaching 100% for the longest excursions. Whether this is the obstruction or a finite-size effect is exactly the question of Part IV.

Part IV

Relating the long shadow windows to the operator gaps

This is the direct test of the keystone arrow: do shadow windows live in sectors where the operator gap is small? The shadow moduli $\{9, 27, 81, 243\}$ are all powers of 3, so multiplicative characters match between Experiments I and F, and the join is on $(Q, \chi\text{-index}, s)$. Because of section 13 the gap is taken from the residue-only operator (F), re-run over these moduli with a *principal-on-units* baseline ($\chi_0 = 1$ on units, 0 on non-units)—the apples-to-apples trivial sector for multiplicative characters.

Tilt level: a clean correspondence

Shadows sit at the weakest-gap tilt

F shows residue-only gaps are systematically weaker (closer to 1) at $s=1$ than at $s=0$. The shadows concentrate there: of 53,158 shadow windows, 43,843 (**82%**) have $\arg\max s=1$, and **all 218 long windows** ($\ell \geq 64$) **are at** $s=1$. Long shadows also sit in weaker-gap sectors: mean gap ratio **0.73** versus 0.57 over all shadows.

Both statements are the manuscript’s predicted direction: a slowly-decaying mode is permitted precisely where the gap is weak, and the critical tilt is where the operator gap is weakest (Target 3 made dynamical).

Sector level: confounded by finite residue-class count

Across sectors at fixed tilt the correspondence is weak, $\text{corr}(|W|, \text{gap}) = +0.247$ (right sign, small). The $\arg\max$ is dominated by the *smallest* modulus: 94% of shadows pick $Q = 9$ (49,727 of 53,158), which has only six unit residues, so a short window’s $|\text{mean } \chi|$ is large by finite-sample inflation regardless of the asymptotic gap (the sector $Q=9, k=3$ carries 36,666 shadows yet has a healthy gap of 0.60).

The finite operator decorrelates every nontrivial sector (all residue-only gaps < 1 , weakest at $s = 1$). The long shadow windows correspond to it in the predicted direction at the level of the tilt and in gap magnitude. But the per-sector single-orbit shadow is dominated by small- Q finite-size inflation, so it is *not* clean evidence of an operator anti-gap. This is the circularity leak made concrete: reading the obstruction off one orbit's short windows is confounded; the operator gap is the reliable object, and it shows no anti-gap.

Part V

Assessment

Where the program stands

Component	Status after testing
Congruence lemma 1	verified, 2.4×10^7 checks [Confirmed]
annealed gap $ \hat{p}(\xi) < 1$	realized; exact operator deviates from it [Refined]
Cramér tilt on critical excursions	bulk leans to $p^{(1)}$, $D(\cdot \ p^{(1)}) \ll D(\cdot \ p)$ [Confirmed]
post-exit taboo	$M_{\text{post}}(1) < 1$, repelled from $p^{(1)}$ [Confirmed]
finite character-sector gap (Target 3)	present, weakest at $s = 1$ (residue-only) [Confirmed]
residue-only uniform κ_Q (Target 1)	stabilizes $\rightarrow 0.236$ ($s=0$)/ 0.074 ($s=1$) over $3^2 \rightarrow 3^7$; leak refuted [Confirmed]
killed-height uniform κ_Q	skew-product = 0 (quadratic coboundary, proved appendix A); factored inherits residue gap [Resolved]
H23a operator vehicle	faithful sync on coboundary-quotient [Decided]
H23a gap (non-coboundary)	constant $\kappa_0 \approx 0.093$ by conductor; autonomy proved, monotonicity open [Reformulated]
shadow $\Rightarrow$ small operator gap	holds by tilt; per-sector confounded by Q [Partial]
Quenched Inverse Principle / keystone	untouched; circularity leak made concrete [Open]

Which operator powers H23a

The proof of appendix A forces a choice of operator, and it resolves cleanly in favour of the **faithful skew product (sync)**, on a *coboundary-quotiented* character family. The reasoning:

1. **Faithfulness is non-negotiable for an inverse argument.** H23a runs *backwards*: a persistent $|W_n(x; \chi, s)|$ must force a near-peripheral eigenstate. That implication is valid only if the operator's spectral radius *is* the growth rate of W_n . Along the real orbit a single valuation b_i drives both the residue step and the tilt weight $2^{s\Delta_{b_i}}$, which is exactly the **sync** coupling. The **factored** operator draws an independent valuation for the height, so its spectral radius does *not* govern W_n .

2. **The factored gap is a mirage.** By corollary 1, $W_n(x; \chi_2, 0) = \frac{1}{n} \sum (-1)^{b_i-1} \rightarrow \mathbb{E}[(-1)^b] = -\frac{1}{3}$ (confirmed: mean -0.353 over 3.2×10^4 orbits). The χ_2 component *does not decay*. The **factored** operator reports a gap there ($\rho < \rho_0$), predicting decay that does not occur—a false negative that would let a genuine non-decaying sector through. **sync** correctly returns $\rho(\mathcal{L}_{\chi_2}) = \rho(\mathcal{L}_1)$ (no decay), matching reality.
3. **The lone ungapped sector is not an obstruction—it is the reference.** χ_2 is a coboundary (theorem 1); its Weyl sum is slaved to the valuation statistics, i.e. to the $\chi = 1$ (height/pressure) data. Equivalently, the orbit’s residues are *not* uniform mod 3: they sit at 1 : 2 ratio $\frac{1}{3} : \frac{2}{3}$, the pushforward of the valuation parity. So the correct annealed residue reference is this non-uniform law, and χ_2 carries no information beyond it. H23a must be posed on the quotient $\widehat{\Omega}_Q / \langle \text{coboundaries} \rangle$, where χ_2 is absorbed into the reference.
4. **On that quotient, sync has a positive, Q -uniform gap—in fact a constant.** The worst non-coboundary character is the fixed conductor-9 (order-6) sector, gap ratio 0.9069 for every $Q \in \{9, 27, 81, 243\}$ and both tilts; higher-conductor characters are strictly more gapped, and each sector’s gap is r -independent because the mod- 3^c reduction is autonomous (proposition 2). Hence

$$\kappa_Q^{(\text{sync, non-coboundary})} = \kappa_0 \approx 0.0931 \quad \text{constant for all } Q = 3^r, r \geq 2$$

(theorem 2)—stronger than the conservative $\gtrsim 1/\text{polylog}(Q)$. It is smaller than the pure residue gap 0.236 (the height-survival mode lifts every sector toward 1) but bounded away from 0, which is all H23a needs.

Decision. Run H23a on the faithful **sync** skew-product operator, relative to the valuation-pushforward residue reference, restricting the character family to the coboundary-quotient. There the finite gap is $\kappa_Q \approx 0.093$, uniform in Q and s . The **factored** and residue-only operators are kept as *diagnostics*: they isolate the pure residue decorrelation ($\kappa = 1 - \frac{2^{1+s}-1}{2^{1+s}+1}$, stabilising at 0.236/0.074) and certify that χ_2 is the unique difference between the two.

Immediate follow-ups. (1) With the operator question settled, the remaining *Sync Quotient Gap Theorem* (theorem 2) has a clean proof program (appendix B): the only missing input is monotonicity in conductor (higher-conductor characters at least as gapped), via complete-fiber principal-unit cancellation; combined with the proved autonomy (proposition 2) this yields the *constant* gap $\kappa_0 \approx 0.093$, with $\kappa_Q \gg 1/\text{polylog}(Q)$ as the fall-back. (2) Switch the defect-window experiment to multiplicative characters for 3^r and measure per-sector $|W_{n,\ell}|$ decay against the F gap rate at fixed (Q, χ, s) , controlling for Q . (3) Normalize the shadow detector by residue-class count so the per-sector signal is not swamped by small Q . (4) Fix the formal C_A -critical certificate so the $B/D/I$ critical labels are canonical.

The object relocates the difficulty rather than removing it. The experiments show $\mathfrak{D}_{\text{Coll}}$ behaves as advertised on the parts that are reorganizations of known structure—the congruence, the annealed gap, the Cramér tilt, the post-exit taboo, and a genuine finite operator gap weakest at the critical tilt—while the one open arrow (a persistent quenched bias forcing a top finite-level eigenstate) remains open, and the single-orbit shadow that would witness it is confounded by finite size exactly as the circularity leak predicts.

A The quadratic-character coboundary: an analytic proof

This appendix proves the structural fact found numerically: for the faithful residue–height skew product, the quadratic character χ_2 of $(\mathbb{Z}/3^r)^\times$ is an exact coboundary, so $\rho(\mathcal{L}_{\chi_2}) = \rho(\mathcal{L}_1)$ and its gap vanishes; whereas every character of order > 2 stays gapped. Write $b_i = v_2(3x_i + 1)$, $B_n = \sum_{i < n} b_i$, and $S(x) = (3x + 1)/2^b$.

Lemma 2 (χ_2 factors through reduction mod 3). *For $r \geq 1$, $(\mathbb{Z}/3^r)^\times$ is cyclic of order $\phi = 2 \cdot 3^{r-1}$ and the quadratic character χ_2 (its unique order-2 character) satisfies $\ker \chi_2 = \{u : u \equiv 1 \pmod{3}\}$. Equivalently $\chi_2(a) = \left(\frac{a}{3}\right)$, the Legendre symbol mod 3: $\chi_2(a) = +1$ if $a \equiv 1$ and -1 if $a \equiv 2 \pmod{3}$.*

Proof. 2 is a primitive root mod 3^r , so the group is cyclic of order ϕ and $\ker \chi_2$ is its unique index-2 subgroup, the squares, of order 3^{r-1} . Reduction $(\mathbb{Z}/3^r)^\times \rightarrow (\mathbb{Z}/3)^\times \cong \mathbb{Z}/2$ has kernel $\{u \equiv 1 \pmod{3}\}$, also of order 3^{r-1} . A cyclic group has a unique subgroup of each order, so the two coincide; hence χ_2 depends only on $a \pmod{3}$, equalling $+1$ on $a \equiv 1$ and -1 on $a \equiv 2$. $\square$

Proposition 1 (Key identity). *For every odd x , with $b = v_2(3x + 1)$,*

$$\chi_2(S(x) \pmod{3^r}) = (-1)^b.$$

In particular the character of the image depends only on b , not on x , and $\chi_2(x_{i+1}) = (-1)^{b_i}$ along any orbit.

Proof. Work mod 3. Since $2 \equiv -1 \pmod{3}$ and $3x + 1 \equiv 1 \pmod{3}$,

$$S(x) = (3x + 1)2^{-b} \equiv 1 \cdot (-1)^{-b} \equiv (-1)^b \pmod{3}.$$

Thus $S(x) \equiv 1 \pmod{3}$ iff b is even and $S(x) \equiv 2 \pmod{3}$ iff b is odd; by lemma 2, $\chi_2(S(x)) = (-1)^b$. (Note $3x + 1 \equiv 1 \pmod{3}$ makes $S(x)$ a unit, so $\chi_2(S(x)) \neq 0$.) $\square$

Corollary 1 (The running character is the valuation parity). $\prod_{i=1}^n \chi_2(x_i) = (-1)^{B_n}$, and $\prod_{i=0}^{n-1} \chi_2(x_i) = \chi_2(x_0)(-1)^{B_{n-1}}$. *The quadratic character accumulated along the orbit is exactly (-1) to the cumulative 2-adic valuation.*

Proof. Immediate from proposition 1: $\prod_{i=1}^n \chi_2(x_i) = \prod_{i=0}^{n-1} (-1)^{b_i} = (-1)^{B_n}$. $\square$

Theorem 1 (χ_2 is a coboundary of the residue–valuation skew product). *On the skew product with state $(a, \tau) \in (\mathbb{Z}/3^r)^\times \times (\mathbb{Z}/2)$ and dynamics $(a, \tau) \xrightarrow{b} ((3a + 1)2^{-b}, \tau + b)$, set $d(a, \tau) = \chi_2(a)(-1)^\tau \in \{\pm 1\}$. Then on every edge*

$$\frac{d(a', \tau')}{d(a, \tau)} = \chi_2(a).$$

Consequently the twisted annealed transfer operator obeys $\mathcal{L}_{\chi_2} = D \mathcal{L}_1 D^{-1}$ with $D = \text{diag}(d)$, so $\rho(\mathcal{L}_{\chi_2}) = \rho(\mathcal{L}_1)$ and $\kappa = 0$, for every tilt s and any height discretization.

Proof. By proposition 1, $\chi_2(a') = (-1)^b$, and $\tau' = \tau + b$ gives $(-1)^{\tau'} = (-1)^\tau (-1)^b = (-1)^\tau \chi_2(a')$. Hence, using $\chi_2(a')^2 = 1$,

$$d(a', \tau') = \chi_2(a')(-1)^{\tau'} = \chi_2(a')(-1)^\tau \chi_2(a') = (-1)^\tau, \quad \frac{d(a', \tau')}{d(a, \tau)} = \frac{(-1)^\tau}{\chi_2(a)(-1)^\tau} = \chi_2(a).$$

With the character carried on the source, the matrix element of $D^{-1}\mathcal{L}_{\chi_2}D$ on this edge is $d(a', \tau')^{-1} w_s(b) \chi_2(a) d(a, \tau) = w_s(b)$, the matrix element of $\mathcal{L}_1$. So the operators are conjugate by the real orthogonal D , hence isospectral. $\square$

Why the residue alone is gapped, and the exact ratio. Without the τ -coordinate, χ_2 is *not* a coboundary: the candidate potential would be a -only, forcing $(-1)^b = 1$. Indeed χ_2 factors through mod 3, on which the dynamics is $\epsilon \mapsto (-1)^b$ on $\{\pm 1\}$; the 2×2 twisted operator $M_{\chi_2}[\epsilon', \epsilon] = \epsilon \sum_{b: (-1)^b = \epsilon'} w_s(b)$ has eigenvalues 0 and $\sum_b (-1)^b w_s(b)$, giving the closed form

$$\frac{\rho(\mathcal{L}_{\chi_2}^{\text{res}})}{\rho(\mathcal{L}_1^{\text{res}})} = \frac{|\sum_b (-1)^b w_s(b)|}{\sum_b w_s(b)} = \frac{2^{1+s} - 1}{2^{1+s} + 1} \quad \left(= \frac{1}{3} \text{ at } s=0, \frac{3}{5} \text{ at } s=1 \right),$$

matching Experiment F exactly. Adding τ supplies the coordinate that absorbs the alternating sign and lifts the ratio to 1.

Remark (Only the quadratic character). For a character χ of order $d > 2$, $\chi(3a+1)$ is *not* constant: by lemma 2 $3a+1$ is a square, and χ restricted to the squares (order 3^{r-1}) is nontrivial when $d > 2$. Then $\chi(a') = \chi(3a+1)\chi(2)^{-b}$ depends on a , the per-edge phase is not a function of (b, τ) alone, and no diagonal D on the skew product trivialises it—so χ stays gapped. This is exactly why the maximum gap ratio over the full group equals 1 only at $k = \phi/2$.

Remark (Height connection; **sync** vs **factored**). The valuation parity is the height's parity: $H_n = n \log_2 3 - B_n + O(1)$, so $\tau = B_n \bmod 2$ is carried by the residue–height state in the **sync** operator (one b drives residue and height together), which therefore inherits the coboundary and shows $\kappa_Q = 0$ for all R, L —as observed. The **factored** operator draws the height's valuation *independently* of the residue's b , so τ no longer records the residue valuation, the coboundary is broken, and the gap returns to the residue value $1 - \frac{2^{1+s}-1}{2^{1+s}+1}$ (and the worst-character κ_Q stabilises). All claims here were checked numerically: the identity of proposition 1 over 10^5 seeds, corollary 1 along every tested orbit, and theorem 1 by direct diagonalisation (residue-only ratio $1/3$; residue $\times$ parity ratio 1).

B The sync quotient gap: a constant, by conductor

This appendix sharpens the remaining target. After quotienting the unique coboundary, the faithful **sync** operator has a gap that is not merely Q -uniform but *constant*, governed by a single fixed finite operator.

Lemma 3 (Coboundary classification). *A character χ of $U_Q = (\mathbb{Z}/3^r)^\times$ is a coboundary of the synchronized residue–valuation skew product iff $\chi \in \langle \chi_2 \rangle = \{1, \chi_2\}$.*

Status. The forward direction is theorem 1. For uniqueness, write $U_{3^r} \cong \{\pm 1\} \times C$ with $C = 1 + 3\mathbb{Z}/3^r$ the principal units (the squares = $\ker \chi_2$); since coboundaries form a group containing χ_2 , it suffices to show no nontrivial character of C is a coboundary. By Wielandt's equality theorem this is equivalent to $\rho(\mathcal{L}_\chi) < \rho(\mathcal{L}_0)$ for every $\chi \notin \langle \chi_2 \rangle$, which the full-character-group computation confirms

for $Q \leq 3^4$ (every $k \neq \phi/2$ has gap ratio < 1). An analytic proof reduces to showing the principal-unit functional equation $\psi(3a+1) = \psi(a^2)$ has no nontrivial solution ψ on C (it already fails at $r = 2$); we leave the all- r statement as lemma 3.

Proposition 2 (Conductor reduction and Q -independence). *For $1 \leq c \leq r$, a character of conductor 3^c depends only on $a \bmod 3^c$, and its synchronized twisted operator is carried by the reduction of the dynamics $\bmod 3^c$. That reduction is autonomous: $(3a+1) \bmod 3^c$ depends only on $a \bmod 3^{c-1}$, so $x_{i+1} \bmod 3^c$ is a function of $(x_i \bmod 3^c, b_i)$ alone. Hence the gap ratio of a conductor- 3^c sector is **independent of r** .*

Proof. $3a \equiv 3(a \bmod 3^{c-1}) \pmod{3^c}$, so $(3a+1)2^{-b} \bmod 3^c$ is determined by $a \bmod 3^c$ and b ; the induced operator on $(\mathbb{Z}/3^c)^\times \times (\text{height})$ makes no reference to r . $\square$

Theorem 2 (Sync quotient gap is constant). *Order the non-coboundary characters by conductor. Numerically, for every $Q = 3^r$ ($r \geq 2$) and every tilt $s \in [0, 1]$, the gap ratio is largest at the smallest conductor 3^2 and decreases with conductor:*

conductor	$3(\chi_2)$	9	27	81	243
order	2	6	9	54	81
gap ratio	1.0000	0.9069	0.8530	0.8261	0.7306

each entry being r -independent (proposition 2). Consequently the worst non-coboundary sector is the fixed conductor-9 (order-6) character, and

$$\kappa_Q^{\text{sync/cob}} = 1 - 0.9069 = \kappa_0 \approx 0.0931 \quad \text{for all } Q = 3^r, \ r \geq 2,$$

a constant, not merely $\gtrsim 1/\text{polylog}(Q)$.

Status. proposition 2 (each conductor's gap is r -independent) is proved; the value κ_0 is the gap of the fixed conductor-9 operator (a small matrix, exactly computable). The remaining input is monotonicity in conductor—that higher-conductor characters are at least as gapped—which the data show cleanly and which is the content of the cancellation step below.

Proof program for the gap theorem.

1. **Coboundary classification** (lemma 3): $\langle \chi_2 \rangle$ is the only character coboundary; reduces to a principal-unit functional equation.
2. **Principal-unit cancellation:** for each non-coboundary χ , a finite admissible block from every bulk state carries untwisted mass $\geq \alpha$ on which $|\mathbb{E}[\chi(\text{endpoint})]| \leq 1 - \eta$. Where the block images contain a full coset of a subgroup H with $\chi|_H \neq 1$, cancellation is *exact* ($\sum_{u \in H} \chi(uh) = 0$)—the likely source of the flat, conductor-monotone gap.
3. **Conductor reduction** (proposition 2): each sector's gap is a fixed r -independent number; the extremal one is conductor 9.
4. **Height/QSD transfer:** the synchronized height weights on the paired cancelling blocks are comparable ($w_s(w_1)/w_s(w_2) \in [C^{-1}, C]$ uniformly for $s \in [0, 1]$) and the blocks stay in the QSD bulk up to $O(e^{-cL})$, so the cancellation survives killing.

Together these give $\kappa_Q \geq \kappa_0 > 0$; the conservative fall-back target is $\kappa_Q \gg 1/\text{polylog}(Q)$.

Remark (What the numerics now mean). The flatness of κ_Q through $Q = 3^5$ and the worst frequency at $\xi = 1/6$ say the extremal obstruction is *bounded-conductor* (conductor 9), not a high-conductor random walk whose gap would decay like Q^{-c} . High-conductor principal-unit characters are killed strongly (apparently by near-complete-fiber averaging), leaving a single fixed low-level sector—hence a true constant gap.